# Jordan L. Shivers

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## Education

**Rice University**, Houston, TX
Ph.D. Candidate, Chemical and Biomolecular Engineering, 2016–2021 (expected)
- Advisor: Fred MacKintosh

**Princeton University**, Princeton, NJ
B.S.E. with Honors (*cum laude*), Chemical and Biological Engineering, 2012–2016
- Certificates (minors) in *Engineering Biology* and *Applications of Computing*
- Advisor: Cliff Brangwynne
- Thesis: Microfluidic immobilization and subcellular imaging of *Caenorhabditis elegans*

## Publications

8. Cell-induced confinement effects in soft tissue mechanics
D. Song, **J. L. Shivers**, F. C. MacKintosh, A. E. Patteson, P. A. Janmey
*Journal of Applied Physics*, (in press, 2021)

7. Compression stiffening of fibrous networks with stiff inclusions
**J. L. Shivers**, J. Feng, A. S. G. van Oosten, H. Levine, P. A. Janmey, F. C. MacKintosh
*Proceedings of the National Academy of Sciences*, **117**: 21037-21044 (2020)

6. Finite size effects in critical fiber networks
S. Arzash, **J. L. Shivers**, F. C. MacKintosh
*Soft Matter*, **16**: 6784-6793 (2020)

5. Nonlinear Poisson effect governed by a mechanical critical transition
**J. L. Shivers**, S. Arzash, F. C. MacKintosh
*Physical Review Letters*, **124**: 038002 (2020)

4. Scaling theory for mechanical critical behavior in fiber networks
**J. L. Shivers**, S. Arzash, A. Sharma, F. C. MacKintosh
*Physical Review Letters*, **122**: 188003 (2019)

3. Stress-stabilized subisostatic rope networks
S. Arzash, **J. L. Shivers**, A. Licup, A. Sharma, F. C. MacKintosh
*Physical Review E*, **99**: 042412 (2019)

2. Normal stress anisotropy and marginal stability in athermal elastic networks
**J. L. Shivers**, J. Feng, A. Sharma, F. C. MacKintosh
*Soft Matter*, **15**: 1666-1675 (2019)

1. Microfluidic immobilization and subcellular imaging of developing *Caenorhabditis elegans*
**J. Shivers**, S. Uppaluri, C. P. Brangwynne
*Microfluidics and Nanofluidics*, **21**: 149 (2017)

## Awards and Honors

2020–21 Lodieska Stockbridge Vaughn Fellowship
*Competitive, university-wide award given to a limited number of students who have demonstrated outstanding achievement and promise*

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| 2020–21 | 1 of 5 Rice nominees for Schmidt Science Fellowship |
| 2020–21 | Sunit Patel ‘85 Endowed Fellowship<br>*Departmental award for research accomplishment* |
| 2020–21 | NASA/Texas Space Grant Consortium Graduate Fellowship |
| 2019–20 | *Honorable mention*, NASA/Texas Space Grant Consortium Graduate Fellowship |
| 2018–19 | Riki Kobayashi Fellowship in Chemical Engineering<br>*Departmental award for best thesis proposal* |
| 2018–19 | Ken Kennedy Institute Oil & Gas High Performance Computing Conference Fellowship |
| 2016–17 | Jack C. Pollard Endowed Fellowship in Engineering |
| 2015–16 | Lidow Senior Thesis Funding |
| 2012–16 | St. Louis–San Francisco Railway–Clark Hungerford ’22 Scholarship in Engineering |
| 2012 | National Merit Scholarship |

## Research Appointments

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| 2016–now | Graduate Fellow, Dept. of Chemical and Biomolecular Engineering<br>**Rice University**, Houston, TX<br>Advisor: Fred MacKintosh |
| 2014–16 | Research Assistant, Dept. of Chemical and Biological Engineering<br>**Princeton University**, Princeton, NJ<br>Advisor: Cliff Brangwynne |
| 2015 | Research Intern, Bioprocess Clinical Manufacturing and Technology<br>**Merck Research Laboratories**, West Point, PA |
| 2014 | Research Intern, Sensors and Integration Group<br>**Applied Science and Technology Research Institute**, Hong Kong |
| 2013–14 | Research Assistant, Dept. of Mechanical and Aerospace Engineering<br>**Princeton University**, Princeton, NJ<br>Advisor: Howard Stone |

## External Courses

**Boulder Summer School for Condensed Matter and Materials Physics**, Boulder, CO
Topic: Theoretical Biophysics, Summer 2019 (3 weeks)

## Invited Talks

*Compression stiffening in biopolymer networks with stiff inclusions*
Rice University ChBE Department, Patel Award Seminar (virtual), Houston, TX (2020)

*Compression stiffening in biopolymer networks with stiff inclusions*
University of Pennsylvania MRSEC IRG2 Weekly Talks (virtual), Philadelphia, PA (2020)

*Mechanical phase transitions and biopolymer network elasticity*
Rice University ChBE Department, Kobayashi Award Seminar, Houston, TX (2019)

## Contributed Talks

*Compression stiffening of fibrous networks with stiff inclusions*
APS March Meeting (virtual, 2021)

*Compression stiffening of fibrous networks with stiff inclusions*
International Congress on Rheology (virtual), Rio de Janeiro, Brazil (2020)

*Nonlinear Poisson effect in critical mechanical networks*
APS March Meeting (virtual), Denver, CO (2020)

*Compression stiffening in fibrous networks with stiff inclusions*
Smalley-Curl Institute Transdisciplinary Symposium, Houston, TX (2020)

*Nonlinear Poisson effect in critical mechanical networks*
Society of Rheology 91st Annual Meeting, Raleigh, NC (2019)

*Scaling theory for mechanical critical behavior in fiber networks*
APS March Meeting, Boston, MA (2019)

*Scaling theory for mechanical critical behavior in fiber networks*
Society of Rheology 90th Annual Meeting, Houston, TX (2018)

*Mechanics of fibrous networks with embedded inclusions*
International Physics of Living Systems Annual Meeting, Houston, TX (2018)

*Anomalous normal stress controlled by marginal stability in fiber networks*
APS March Meeting, Los Angeles, CA (2018)

**Contributed Posters**

*Nonlinear Poisson effect in critical mechanical networks*
Gordon Research Conference on Soft Condensed Matter Physics, New London, NH (2019)

*Nonlinear Poisson effect in critical mechanical networks*
Boulder Summer School for Condensed Matter and Materials Physics, Boulder, CO (2019)

*Nonlinear Poisson effect in critical mechanical networks*
International Soft Matter Conference, Edinburgh, UK (2019)

*Nonlinear Poisson effect in critical mechanical networks*
APS March Meeting, Boston, MA (2019)

**Community/Outreach**

**Peer review** for *Physical Review Letters*, *Biophysical Journal*, *Soft Matter*, and *Macromolecules*
**Volunteer physics tutor**, Rice NEWT K-12 Tutoring Program, 2021-present
**Graduate recruitment co-chair** for Rice Chemical and Biomolecular Engineering GSA, 2017-18
**Graduate fellow**, Duncan College (undergraduate residential college), 2017-19
**Dean's *Engineering Student Advisory Council***, 2017-18
**Presentation judge**, Gulf Coast Undergraduate Research Symposium, 2017
**Volunteer admissions interviewer**, Princeton Alumni Schools Committee, 2016-present
**Instructor**, Princeton Splash, 2016

**Professional Associations**

Society of Rheology
American Physical Society

**Teaching**

**TA, Rice University**
CHBE 603: Rheology, Spring 2019
CHBE 402: Transport Phenomena II, Spring 2017
CHBE 401: Transport Phenomena I, Fall 2017
CHBE 403: Design Fundamentals, Fall 2016

**References**

**Fred C. MacKintosh, Ph.D.**
Abercrombie Professor of Chemical and Biomolecular Engineering
Professor of Chemistry, Physics and Astronomy
Rice University
fcmack@rice.edu

**Paul A. Janmey, Ph.D.**
Professor, Institute for Medicine and Engineering and Dept. of Physiology
University of Pennsylvania
janmey@mail.med.upenn.edu

**Herbert Levine, Ph.D.**
Distinguished Professor, Depts. of Physics and Bioengineering
Northeastern University
h.levine@northeastern.edu